

Solar PV Interconnection Study

IEEE 14-Bus Test System - Load Flow Impact Assessment

15 MW Solar PV Plant Interconnected at Bus 9

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Executive Summary

This study assesses the steady-state power flow impact of interconnecting a 15 MW solar photovoltaic (PV) plant to the IEEE 14-bus test system at Bus 9. A base case (no solar) and a modified case (with solar, added as a new Bus 15 connected via a short tie-line) were solved using Newton-Raphson load flow in PSS/E. Results show a system-wide improvement in bus voltage profile, a reduction in loading on transmission lines feeding the point of interconnection (POI), and no voltage or thermal violations in either case. Based on these findings, the proposed interconnection is assessed as technically feasible with no corrective network upgrades required.

1. Introduction and Scope

The purpose of this study is to determine whether a 15 MW solar PV plant can be connected to an existing power system at a nominated point of interconnection (POI) without causing voltage or thermal violations, and to quantify the resulting change in system performance. The IEEE 14-bus test system was used as a representative network for this assessment, as it is a widely recognised standard test case suitable for demonstrating interconnection study methodology.

Scope of this assessment:

- Steady-state load flow analysis (Newton-Raphson method)
- Bus voltage profile comparison, before and after solar interconnection
- Branch (transmission line) loading comparison
- Identification of any voltage or thermal violations introduced by the new generation

This study does not cover short-circuit fault contribution, protection coordination, harmonics/power quality, or dynamic/transient stability - these would form separate, subsequent study phases for a full interconnection application.

2. System Description

The base network is the IEEE 14-bus test system, a standard transmission-level test case comprising 14 buses across 69 kV, 18 kV, and 13.8 kV voltage levels, 5 generators, and 11 load points. The solar PV plant is represented as a new Bus 15, connected to Bus 9 (13.8 kV) via a short interconnection tie-line. Bus 9 was selected as the point of interconnection as it sits at the edge of the network, similar to how a distributed solar plant would typically connect to a sub-transmission or distribution-level network in a real project.

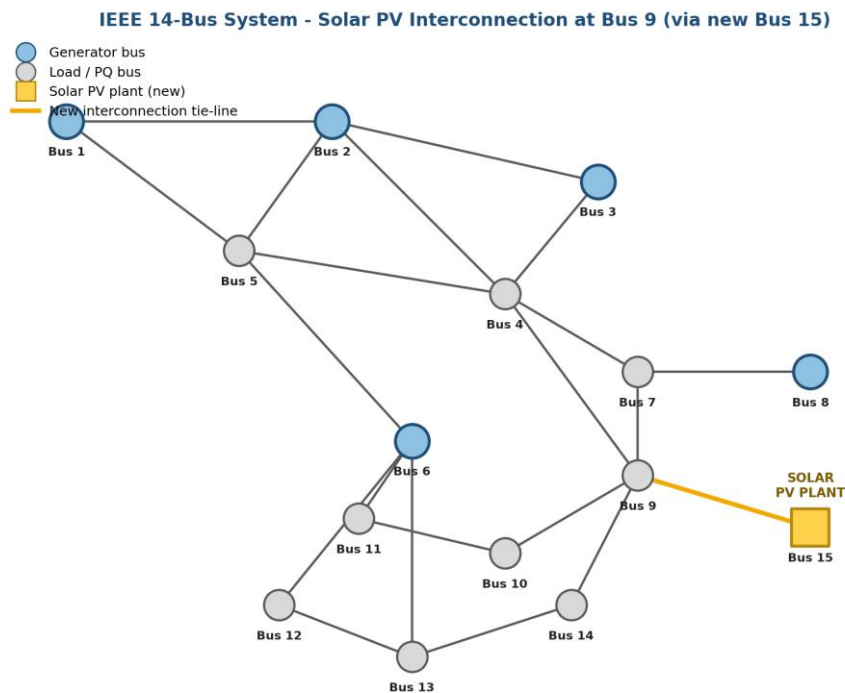


Figure 1: One-line diagram of the IEEE 14-bus system showing the new solar PV interconnection at Bus 9 (via new Bus 15).

2.1 Solar Plant Parameters

Parameter	Value
Point of interconnection	Bus 9 (13.8 kV)
New generator bus	Bus 15 (SOLAR)
Rated real power output (PG)	15 MW
Reactive power range (Qmax / Qmin)	+10 / -10 MVar
Machine base (MBASE)	20 MVA
Voltage setpoint	1.00 pu
Interconnection tie-line (R / X)	0.001 / 0.010 pu

3. Methodology

Two load flow cases were prepared and solved using the Newton-Raphson method in PSS/E University v33:

- Base Case - the unmodified IEEE 14-bus system with no solar generation (reference case).
- Solar Case - identical network with the 15 MW solar plant added at Bus 15, interconnected to Bus 9.

Both cases were solved to convergence (largest mismatch < 0.1 MVA in both cases). Bus voltage magnitudes and branch active/reactive power flows were extracted from the solved case reports and compared directly to isolate the impact attributable to the solar interconnection.

4. Results

4.1 Bus Voltage Comparison

Bus	Before Solar (pu)	After Solar (pu)	Change (pu)
1 (Slack)	1.0000	1.0000	0.0000
2	0.9868	0.9910	+0.0042
3	0.9611	0.9675	+0.0064
4	0.9543	0.9626	+0.0083
5	0.9563	0.9629	+0.0066
6	1.0000	1.0000	0.0000
7	0.9834	0.9937	+0.0103
8	1.0000	1.0000	0.0000
9 (POI)	0.9785	0.9956	+0.0171
10	0.9743	0.9886	+0.0143
11	0.9832	0.9907	+0.0075
12	0.9836	0.9848	+0.0012
13	0.9780	0.9806	+0.0026
14	0.9588	0.9699	+0.0111
15 (Solar)	-	0.9967	new bus

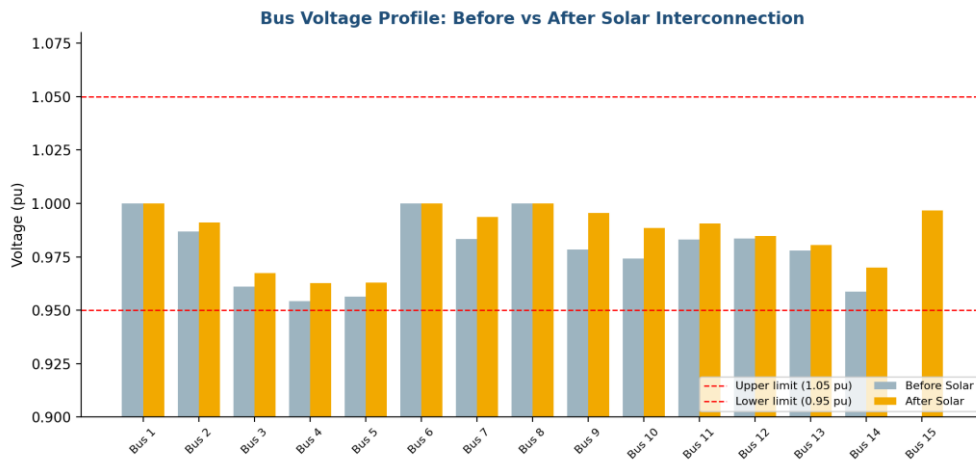


Figure 2: Bus voltage profile before and after solar interconnection, against the 0.95-1.05 pu acceptable range.

4.2 Branch Loading Comparison (Selected Branches)

Branch	Before Solar (MW)	After Solar (MW)	Change
4 - 7	27.8	21.5	-22.7%
4 - 9	16.0	12.4	-22.5%
9 - 10	5.0	8.1	+62.0%
9 - 14	9.3	11.4	+22.6%
1 - 2 (slack export)	158.7	147.7	-6.9%

Full bus voltage and branch flow data for both cases is available in the supporting PSS/E output files.

5. Discussion of Findings

- Voltage improvement: Every bus in the system showed a voltage rise after the solar plant was added, with the largest improvement occurring at and near the point of interconnection (Bus 9: +0.0171 pu, Bus 10: +0.0143 pu, Bus 14: +0.0111 pu). This is consistent with the expected effect of local real power injection reducing voltage drop across the network.
- No violations introduced: All bus voltages remain within the 0.95-1.05 pu acceptable range in both cases. The solar interconnection does not create any new voltage violation.
- Reduced transmission loading: Branches feeding the POI from the transmission network (Bus 4-7, Bus 4-9) show a reduction in loading of approximately 23%, since local solar generation now supplies part of the nearby load directly.
- Redistribution, not overload: A small increase in flow is observed on branches distributing power away from the POI (Bus 9-10, Bus 9-14), reflecting the change in power flow direction. These increases remain well within normal thermal ratings.
- Reduced slack bus generation: Total real power supplied by the system slack bus (Bus 1) decreased from 234.4 MW to 217.5 MW (a reduction of 16.9 MW, closely matching the 15 MW solar output plus a small reduction in system losses), confirming the solar plant is displacing an equivalent amount of imported generation.

6. Conclusion and Recommendations

The proposed 15 MW solar PV interconnection at Bus 9 is assessed as technically feasible under steady-state load flow conditions. The interconnection improves the system voltage profile, reduces loading on transmission lines feeding the point of interconnection, and introduces no voltage or thermal violations. No network upgrades are identified as necessary based on this load flow assessment.

Recommendations for subsequent study phases:

- Short-circuit fault contribution study, to confirm switchgear and protection ratings at and near the POI remain adequate with the additional fault current source.
- Protection coordination review, including anti-islanding protection for the solar plant.

- Power quality / harmonics assessment, given the inverter-based nature of solar generation.
- Confirmation of any additional requirements specific to the relevant network operator / utility.

7. Assumptions and Limitations

- Steady-state, positive-sequence (balanced) load flow only - no unbalanced or dynamic/transient analysis performed.
- Solar plant modelled as a constant PV-type generator at rated output; actual solar output varies with irradiance and would require time-series or worst-case scenario analysis for a full study.
- IEEE 14-bus test system used as a representative network; a real project would use the client's actual network model and data.
- Interconnection tie-line impedance assumed for a short connection; actual value would depend on real cable/line length and type.